B.E. EVEN Mid-Semester Examinations 2024-25

Subject: Environmental Science

Code: MC-HU 402

Stream: IT | **Time:** 1 Hour | **Full Marks:** 20

1. Define sustainable development. How EIA can help in achieve sustainable development? (4)

Answer:

Sustainable Development: It is defined as development that meets the needs of the present generation without compromising the ability of future generations to meet their own needs. It requires balancing economic growth, environmental protection, and social equity.

Role of EIA in Sustainable Development:

Environmental Impact Assessment (EIA) is a formal process used to predict the environmental consequences of a proposed project before it proceeds.

- **Preventive Action:** It identifies potential ecological damages early, allowing developers to modify plans or adopt cleaner technologies.
- **Resource Management:** Ensures optimal and sustainable use of natural resources.
- **Mitigation:** Recommends mitigation strategies (like afforestation or waste treatment) to minimize adverse effects, thereby aligning industrial/economic growth with long-term ecological stability.

2. a) What are the basic types of ecosystem? (2)

Answer:

The basic types of ecosystems are broadly classified into two categories:

- **Terrestrial Ecosystems:** Land-based ecosystems. Examples include Forests, Grasslands, Deserts, and Tundras.
- **Aquatic Ecosystems:** Water-based ecosystems. Examples include Freshwater ecosystems (lakes, rivers, ponds) and Marine ecosystems (oceans, coral reefs, estuaries).

2. b) Distinguish between abiotic and biotic components of an ecosystem. (3)

Answer:

Feature	Abiotic Components	Biotic Components
Definition	Non-living physical and chemical elements in the ecosystem.	Living organisms that shape the ecosystem.
Examples	Sunlight, temperature, water, soil minerals, air, humidity.	Plants (producers), animals (consumers), bacteria/fungi (decomposers).

Role	They establish the environment and determine which organisms can survive in it.	They interact with each other and the abiotic environment for energy flow and nutrient cycling.

3. Describe energy flow model with the help of the laws of thermodynamics. (5)

Answer:

The flow of energy in an ecosystem is unidirectional and is strictly governed by the two laws of thermodynamics:

- **First Law of Thermodynamics (Conservation of Energy):** Energy can neither be created nor destroyed; it can only be transformed from one form to another.
  - *In Ecosystems:* Solar energy is captured by producers (plants) and transformed into chemical energy (food) via photosynthesis. This energy is then transferred sequentially through the food chain to herbivores, carnivores, and decomposers. Total energy remains constant in the broader universe.
- **Second Law of Thermodynamics (Law of Entropy):** No energy transfer is 100% efficient. With every transformation, some usable energy is dissipated, usually as heat.
  - *In Ecosystems (The 10% Rule):* As energy flows from one trophic level to the next (e.g., plant to deer, deer to tiger), about 90% of it is lost for metabolic processes (respiration, movement) and as heat. Only about 10% of the energy is stored as biomass in the next level. This explains why food chains rarely exceed 4 or 5 trophic levels, as the available energy eventually becomes too small to sustain life.

4. a) Define green belt. (2)

Answer:

A **Green Belt** refers to an area of open land, forests, parks, or agricultural spaces intentionally preserved around urban, industrial, or residential areas. Building and heavy development are heavily restricted here to prevent urban sprawl, improve air quality (acting as a sink for pollutants), and provide natural habitats.

4. b) Describe the controlling measures of noise pollution. (3)

Answer:

Controlling measures for noise pollution involve interventions at different stages:

- **Control at Source:** Using silencers/mufflers in vehicles and industrial generators, regular lubrication of machinery to reduce friction, and designing quieter engines.
- **Control in the Transmission Path:** Planting dense green belts (trees act as acoustic barriers), using soundproofing materials in building walls, and constructing sound barriers alongside highways.
- **Control at Receiver End:** Using personal protective equipment (PPE) like earplugs or earmuffs for industrial workers exposed to high decibels.
- **Legislative Measures:** Strictly enforcing zoning laws to keep industrial zones away from residential areas and restricting the use of loudspeakers during nighttime.

5. Explain how particulate matter (PM) contributes in air pollution. (5)

Answer:

Particulate Matter (PM) refers to microscopic solid or liquid particles suspended in the air. PM is a major contributor to air pollution due to its diverse sources and severe impacts:

- **Sources:** Produced naturally by dust storms and forest fires, and anthropogenically by vehicle exhausts, industrial smokestacks, construction dust, and coal burning.
- **Classification:** Grouped by size, primarily **PM10** (inhalable particles < 10 micrometers) and **PM2.5** (fine inhalable particles < 2.5 micrometers).
- **Contribution to Pollution:**
    - **Health Impacts:** Due to their minuscule size, PM2.5 can bypass the nose and throat, penetrating deep into the lungs and even entering the bloodstream. This causes asthma, bronchitis, cardiovascular diseases, and premature death.
    - **Visibility Reduction (Smog):** PM absorbs and scatters sunlight, severely reducing visibility, which causes haziness and operational issues for aviation and transport.
    - **Environmental Damage:** PM can settle on soil and water bodies, altering their nutrient/chemical balance (making lakes acidic). It also settles on plant leaves, blocking sunlight and hindering photosynthesis.